Francesco Rosa, Ph.D.

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SUMMARY

Computer Engineer specialized in Artificial Intelligence, **Physical AI**, Robotics, **Computer Vision**, **Human-Robot Interaction**, Robot Manipulation, and Pattern Recognition.

EDUCATION

University of Salerno
Ph.D. Information Engineering
Excellent Nov. 2021– Feb. 2025
Thesis: "Video-Conditioned Multi-Task Imitation Learning for Robotic Systems: Enhancing Robustness Through Object-Centric Reasoning"

University of Salerno
M.Sc. Computer Engineering
110/110 Cum Laude Sep. 2019 - Oct. 2021
Thesis: "3D Deep Neural Network for the 6DoF Grasp Pose Estimation on an Embedded Platform"

CORE SKILLS

Programming
Python, C/C++, CUDA

ML & Vision
PyTorch, Torch Geometric, TensorFlow, Hugging Face, OpenCV, Scikit-Learn

Robotics
ROS/ROS2, MoveIt, MPC, SLAM, Planning&Navigation, Robosuite, MuJoCo, Nvidia Isaac Sim, URDF

Hardware
Universal Robots, Nvidia Jetson, Intel RealSense, ZED, Docker, Git

Relevant Projects

Real-World Deployment Human-Video Conditioning

Vision-Language-Action Models Benchmark

Video-Conditioned Imitation Learning

6DoF Grasp Pose Estimation

UR10 Remote Control with Haptic Interface

Planning Parameters Optimization for Turtlebot mobile robot

Optimization Techniques for Trajectory Planning

PROFESSIONAL EXPERIENCE

Postdoc Researcher
University of Salerno — *Jan. 2026 – Present, Salerno, Italy*

- **Cross-Embodiment Learning**: Designing Physical-AI Models for robotics to enable zero-shot task transfer from human video demonstrations to robotic platforms. Analysis with both simulation and real-world environments.

Contract Researcher
University of Salerno — *Sept. 2025 – Jan. 2026, Salerno, Italy*

- **VLA Architectures**: Research on data-driven control policies integrating Vision-Language-Action Models. Analysis of spatial and skill-combination zero-shot generalization capabilities.

Visiting Ph.D. Student
ENSICAEN - GREYC Lab — *Apr. 2024 – Jul. 2024, Caen, France*

- **Graph Neural Networks**: Researched GNN-based heuristic estimators for symbolic task planning, under the supervision of Prof. *Luc Brun*.

Ph.D. Candidate
University of Salerno — *Nov. 2021 – Feb. 2025, Salerno, Italy*

- **Imitation Learning**: Designed object-centric deep learning architectures for multi-task manipulation. Improved success rates by **55%**.
- **Real-World Deployment**: Validated algorithms on physical UR5e robot.
- **Data**: Collected both simulated and real-world dataset.

Tutor & External Expert
University of Salerno — *Nov. 2021 – Sept. 2025, Salerno, Italy*

- **Undergraduate courses**: Conducted exercise lessons for Programming Fundamentals, Algorithms and Data Structures, Computer Systems, Physics 2, and General Physics.
- **Training courses**: Delivered training courses for external organizations on Machine Learning, and Computer Vision for Industrial Robotics.

R&D Engineer & Co-founder
Kalya — *Jan. 2020 – Dec. 2021, Avellino, Italy*

- **Embedded Systems**: Designed and implemented the image acquisition and processing system for the smart trap *Sentinella*, based on ESP-32 microcontroller.

EUROPEAN RESEARCH PROJECTS

Reseacher
H2020 - "FELICE" ,

- **Human-Robot Collaboration**: In collaboration with FIAT Research Center, and international organizations, developed a multimodal command classification interface (voice/gesture) on **Nvidia Jetson Xavier**, for a mobile robotic platform.

HONORS & AWARDS

- **National Innovation Award (PNI) 2020**: Finalist in CleanTech category (Top startups in Italy).
- **Start-Cup Campania**: 2nd Place with project Sentinella.